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Cortical network dynamics and neural decoding of fine motor complexity via fNIRS and attention-based deep learning

Xing Ji [a], Zhong Yin [a], Yifei Bi [b], Kaiwei Yu [a], Yize Li [a], Jiafa Chen [a,*], Dawei Zhang [a]

[a] Engineering Research Center of Optical Instrument and System, Ministry of Education and Shanghai Key Lab of Modern Optical System, University of Shanghai for Science and Technology, Shanghai, China
[b] College of Foreign Languages, University of Shanghai for Science and Technology, Shanghai, China

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ABSTRACT

Fine motor decline serves as a critical early biomarker for neurodegenerative diseases like Parkinson's disease, making its accurate assessment essential for early detection and intervention. While functional near-infrared spectroscopy (fNIRS) offers a portable, non-invasive neuroimaging solution, the precise cortical dynamics underlying varying levels of motor complexity remain underexplored. This study aims to investigate how fine motor task complexity modulates cortical activation and functional network topology. A secondary objective is to develop and validate a high-performance deep learning model to classify motor complexity levels from fNIRS signals. fNIRS data were recorded from healthy participants performing five fine-motor tasks of increasing complexity, and activation analyses were combined with graph-theoretical metrics to characterize neurophysiological responses. To classify the complexity of fine motor tasks from fNIRS signals, this study developed a bidirectional long short-term memory (Bi-LSTM) model. Performance evaluation used leave-one-out cross-validation, supplemented by multi-seed training to improve robustness. The model achieved an average classification accuracy of 90.67% $\pm$ 7.07% (95% CI: $\pm$ 2.68%) and an AUC of 0.9720 $\pm$ 0.0431, outperforming traditional support vector machine (by 21.3%) and Bi-LSTM (by 10.97%). These results demonstrate the model's strong generalization across subjects and its ability to capture temporal-spatial patterns of cortical activation associated with increasing task complexity, providing a promising foundation for fine motor decoding and adaptive neurorehabilitation.

1. Introduction

Neurodegenerative disorders, particularly Parkinson's disease, have become significant global health challenges, with rising prevalence linked to aging populations and escalating healthcare demands[1]. Among early clinical signs, impairments in fine-motor function have been identified as sensitive indicators of disease onset [2,3]. These deficits not only compromise patients' autonomy and quality of life; they also impose substantial economic burdens on families and society. Notably, motor dysfunction may precede cognitive decline by up to a decade [4], highlighting the potential of fine-motor assessments as prodromal biomarkers for neurodegenerative progression. Consequently, the development of sensitive and quantitative evaluation tools is critical for early diagnosis, longitudinal monitoring, and targeted therapeutic interventions.

Current motor-assessment methodologies span behavioral evaluations, peripheral physiological measurements [5,6], and neuroimaging techniques. Conventional clinical scales, such as the Movement Disorder Society-Unified Parkinson's Disease Rating Scale[7] and the Fugl-Meyer Assessment [8] remain widely used but suffer from inter-rater variability and limited sensitivity to subtle motor deficits. To address these limitations and to improve the accuracy of motor function assessment, more objective physiological techniques have been employed. Notably, Bisdorff et al. (1999) employed electromyography (EMG) to examine neuromuscular responses during free fall in both healthy individuals and patients with motor impairments. This validated the utility of EMG in detecting neuromuscular dysfunction [9]. Subsequent research used EMG to characterize lower-limb muscle activity in individuals with Parkinson's disease during gait tasks, providing quantitative metrics of motor dysfunction [10].

Despite their utility, however, EMG and other peripheral physiological signals offer limited insight into the central neural mechanisms

* Corresponding author.
E-mail address: jiafachen@usst.edu.cn (J. Chen).

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underlying motor impairment, particularly those involving cortical processes. In contrast, neuroimaging modalities have achieved prominence through their ability to monitor cortical dynamics associated with fine-motor decline directly. Nonetheless, conventional neuroimaging techniques, such as electroencephalography (EEG) and functional magnetic resonance imaging (fMRI), present critical drawbacks. Notably, EEG is highly susceptible to motion artifacts, while fMRI is poorly suited for naturalistic movement paradigms. Functional near-infrared spectroscopy (fNIRS) has thus emerged as a compelling alternative, offering portability, resilience to motion artifacts, and robustness against environmental noise [11–13]. In fact, fNIRS infers neural activity by tracking changes in oxygenated (HbO) and deoxygenated (HbR) hemoglobin levels in cortical gray matter, leveraging neurovascular coupling as an indirect yet robust mechanism [14].

To date, fNIRS studies have primarily focused on gross motor functions, particularly upper- and lower-limb movements [15,16]. In contrast, investigations into fine-motor control remain relatively scarce, and these have primarily been limited to simple tasks such as single-finger tapping [17] or precision grasping [18], without considering the complexity dimension. Task complexity provides a quantifiable framework for probing the integrity of motor control systems, enabling early detection of functional decline in conditions such as Parkinson's disease. However, the neural correlates of graded variations in fine-motor task complexity remain underexplored, particularly with respect to their potential to serve as biomarkers of motor dysfunction. Clarifying how cortical dynamics respond to increasing task demands is crucial for understanding early-stage motor deficits.

While fNIRS shows potential for classifying fine-motor behaviors, its capacity to quantitatively model neuroplastic adaptations across varying levels of task complexity remains largely unexplored. In neural engineering and rehabilitation applications, accurate classification of fine hand movements is essential for both elucidating motor control mechanisms and enabling naturalistic brain-computer interface (BCI) interaction. Traditional machine-learning approaches have been extensively applied to the decoding of motor intent. Support vector machines (SVM) offer robust performance on small datasets [19]. Meanwhile, random forests (RF) can effectively handle multivariate inputs and evaluate feature importance, making them suitable for coarse-level motor classification [20]. Nevertheless, these conventional models often exhibit suboptimal performance when processing high-dimensional temporal data, and they lack the capacity to capture long-range dependencies.

Recent advances in deep learning have broadened the methodological toolkit for fNIRS-based analysis. Convolutional neural networks are commonly employed to extract spatial features from multi-channel signals, while recurrent models, such as long short-term memory, effectively capture temporal dependencies in hemodynamic responses [21]. More recently, Transformer-based models have gained attention for their ability to model long-range dependencies via self-attention. However, their data-intensive nature limits their applicability to small-sample fNIRS datasets [22,23]. Generative adversarial networks have also been explored for data augmentation and cross-subject generalization [24]. These approaches reflect a shift toward data-driven modeling in fNIRS research, yet model selection must balance performance, interpretability, and robustness in low-SNR, limited-data settings. In such contexts, bidirectional long short-term memory (Bi-LSTM) networks offer a promising alternative, providing temporal modeling capabilities with relatively lower data demands and enhanced robustness. These features make Bi-LSTM particularly suitable for decoding complex, dynamic fine-motor behaviors from limited fNIRS samples.

Leveraging fNIRS-derived hemodynamic responses and a structured fine-motor task paradigm, this study deploys graph-theoretical methods to explore the topological properties of cortical networks. By examining how task complexity modulates cortical activation and connectivity, this study provides insights into neural-resource reallocation under varying cognitive and motor demands. In parallel, an attention-based Bi-LSTM classification framework was implemented to evaluate the capacity of fNIRS signals to decode fine-motor behaviors across different levels of complexity. Together, these findings establish a unified framework for probing the cortical mechanisms of fine-motor control and for enhancing task-complexity decoding in neuroengineering applications.

2. Methods

2.1. Participants

Fifteen healthy, right-handed university students (assessed via the Edinburgh Handedness Inventory) were recruited from the University of Shanghai for Science and Technology for participation in this study. The sample comprised eight females (mean age = 24.3 ± 2.29 years; range: 22–29) and seven males (mean age = 23.5 ± 0.96 years; range: 22–25). In addition, given the relatively modest sample size (n = 15), we conducted post hoc power analyses using G*Power 3.1.9 based on the repeated-measures ANOVA with task complexity as the main effect, to evaluate the sensitivity of the current sample size to detect complexity-related effects. All participants had normal or corrected-to-normal vision, with no history of neurological or psychiatric disorders, and no color-vision deficiencies. Before the experiment, participants were fully informed about the study procedures, task protocols, and safety considerations. They were informed of their right to withdraw at any point without penalty. The study protocol adhered to the Declaration of Helsinki and the International Ethical Guidelines for Health-Related Research Involving Humans, and it received approval from the Medical Ethics Committee of Gongli Hospital, Pudong New District, Shanghai. Written informed consent was obtained from all participants before they participated in the study.

2.2. Experimental environment and paradigm

Participants were seated 80–90 cm from a display screen in a quiet, softly lit room, with both hands resting on a table to minimize motion artifacts. Visual instructions guided them through five fine-motor tasks in a fixed sequence: fist clenching, hand flipping, paper tearing, bottle cap twisting, and using chopsticks to pick up soybeans (chopsticks-soybean). The tasks were designed to increase in difficulty to avoid psychological strain gradually. Based on eye coordination, precision demand, force exertion, and muscular engagement, the tasks were classified into low, medium, or high complexity. Fist clenching and hand flipping were rated as low, paper tearing and bottle cap twisting as medium, and chopsticks-soybean as high complexity. This classification was validated by subjective ratings from 100 participants (see Section 3.1).

The experimental procedure followed the paradigm depicted in Fig. 1. Participants remained seated in a resting state for 30 s before task initiation. A fixation cross ("+") appeared on the screen for two seconds to signal task onset. Subsequently, participants completed the five tasks in a fixed order, guided by on-screen cues. Each task was performed for five seconds, followed by a 10-second rest interval. A single trial included all five tasks (75 s), and the experiment ended after two trials, followed by a final 30-second rest.

2.3. Data acquisition

Functional brain activity was recorded using the Brite MKIII portable fNIRS system (Artinis, Netherlands), which measures task-evoked changes in HbO and HbR concentrations. As shown in Fig. 2, the system comprises a wearable cap equipped with light sources, photo detectors, and a control unit, with data acquired at a sampling rate of 50 Hz. The optode layout included 10 dual-wavelength light sources (760/850 nm) and eight detectors, with a 30 mm source-detector spacing. In accordance with the international 10–20 system, fNIRS channels were arranged over the primary motor cortex (M1) and premotor cortex

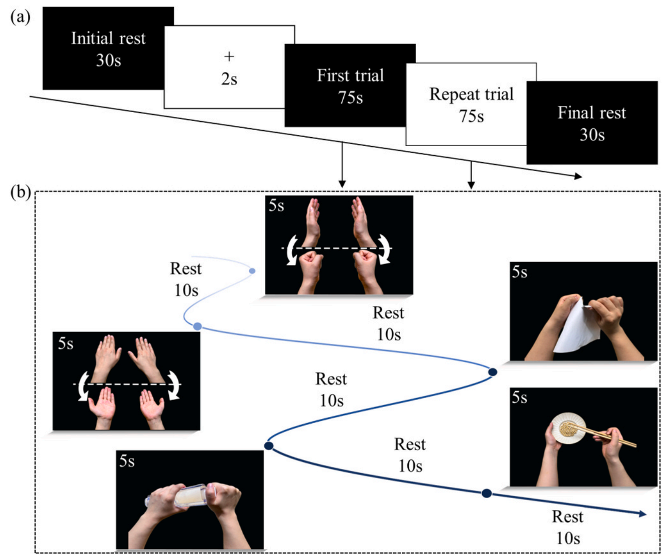

Fig. 1. Experimental paradigm. (a) Overall temporal structure of the experiment. (b) Temporal sequence of a single trial.

(PMC) using a predefined optode montage. These regions are functionally involved in the execution and planning of fine-motor tasks [25,26]. A total of 24 measurement channels were established (Fig. 2f). The layout of the experimental environment is shown in Fig. 2g.

2.4. Data preprocessing

This study preprocessed fNIRS data using the Homer2 toolbox in MATLAB (2021) [27] (Fig. 3). First, channel quality was automatically screened using EnPruneChannel: channels were deemed low-quality and discarded when the average optical intensity fell outside the acceptable range of 0.01 to 1, or when the signal-to-noise ratio (SNR) was < 2 (at SNR < 2, background noise and valid blood flow signals are at comparable levels, rendering the channel unreliable for reflecting task-related hemodynamic changes) [28,29]. This experiment implemented rigorous online quality control prior to acquisition (including probe-position adjustment, optical-path clearance verification, and real-time optical-intensity monitoring). Consequently, no channels were discarded for any subject, ensuring overall high data quality. Subsequently, hmrIntensity2OD was used to convert raw optical intensities into optical density (Δ OD) signals.

Since task-related hemodynamic changes manifest as gradual shifts,

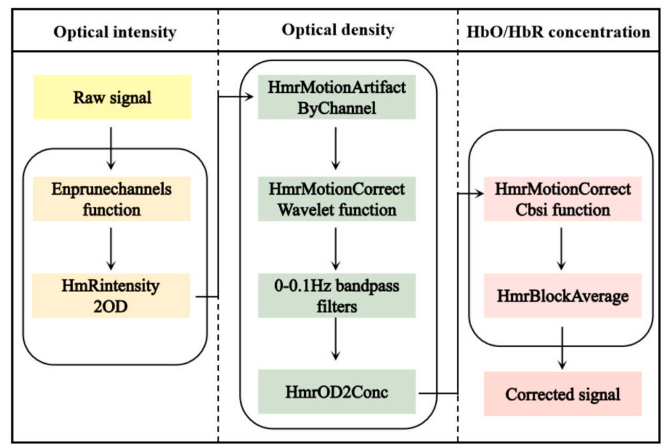

Fig. 3. Data preprocessing flowchart.

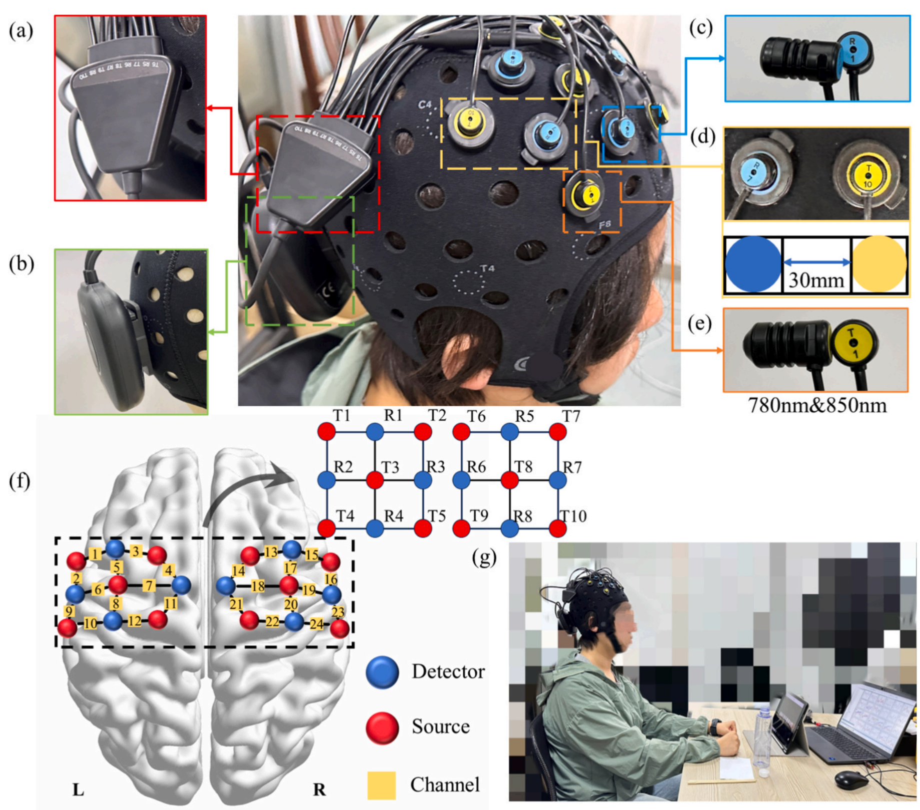

Fig. 2. Experimental setup. (a)Optical collector, (b) Host module, (c) Photodetector, (d) Source-detector distance (30 mm), (e) Dual-wavelength light source (760 nm and 850 nm), (f) Channel distribution, (g) Layout of the experimental environment.

while motion artifacts appear as brief, sudden spikes, we employed hmrMotionArtifactByChannel to detect and mark short-term spike segments based on a defined time window (tMotion = 0.5 s), extended duration (tMask = 1 s), and amplitude thresholds (std_thresh = 5, amp_thresh = 0.05) to detect and mark short-term abrupt segments[30]. Subsequently, wavelet correction was performed using hmrMotionCorrectWavelet. This method employs translation-invariant wavelet decomposition to suppress anomalous wavelet coefficients and reconstruct the signal, thereby removing artifacts while preserving task-related gradual components. In this study, the Daubechies-2 wavelet was used with four levels of decomposition and a threshold coefficient of iqr = 0.1.

To further reduce physiological noise such as heartbeat (1–1.5 Hz), respiration (0.2–0.5 Hz), and blood pressure fluctuations (approximately 0.1 Hz), BandPassFilters were applied to implement a 0–0.1 Hz bandpass filter. Subsequently, hmr_OD2Conc converted Δ OD into HbO and HbR concentration changes based on the modified Beer–Lambert law. To enhance signal quality, hmrMotionCorrectCbsi further applied correlation enhancement correction based on the negative correlation between HbO and HbR. Finally, HmrBlockAverage extracts the task-related hemodynamic response. Given that HbO is more sensitive to changes in neural activity [31], this study focuses on analyzing the HbO signal.

2.5. Graph-theoretical analysis of brain networks

Graph-theoretical analysis was performed on the fNIRS data to quantify the manner in which topological properties of brain functional networks change with varying complexity during fine-motor task execution. Each of the 24 fNIRS channels was treated as a network node, and the pairwise Pearson correlation coefficients of the HbO signals were computed to define edge weights, resulting in a 24 × 24 functional connectivity matrix [3,32–34]. Each trial contained five task segments, and for each task, a corresponding five-second HbO window was extracted to construct a functional connectivity matrix. To eliminate weak or statistically insignificant connections, a sparsity-thresholding approach was applied, converting the weighted functional connectivity matrix into a binary adjacency matrix [35]. In this study, the five global network features, namely, clustering coefficient(C_p), characteristic path length(L_p), global efficiency(E_g), local efficiency(E_{loc}), and connection strength(C_d), were examined. Among these, C_p and E_{loc} evaluated the aggregation of the network, whereas L_p and E_g evaluated the information transmission efficiency of the network. Meanwhile, C_d indicated global connectivity across cortical regions. The formal definition is as follows:

C_p reflects the degree of coordination among different brain regions during tasks, a higher C_p value indicates that the cortical network can rapidly and efficiently integrate information across regions to support the execution of more complex fine-motor actions.

$$C_p = \frac{2E_v}{k_v(k_v-1)}, \tag{1}$$

where E_v represents the set of edges connected to node v, and k_v denotes its degree. The L_p describes the distance information travels between brain regions. In more complex actions, the cortex requires more frequent communication between regions to maintain precision, and the L_p will decrease accordingly.

$$L_p = \frac{1}{N(N-1)}\sum_{i\neq j} d_{ij}, \tag{2}$$

where N denotes the total number of nodes, and d_{ij} denotes the shortest path length between nodes i and j. E_g measures the efficiency of information transmission across the entire brain network during tasks, higher E_g indicates that neural information can be transmitted more rapidly and smoothly among distributed cortical regions, facilitating more coordinated motor control.

$$E_g = \frac{1}{N(N-1)}\sum_{i\neq j}\frac{1}{d_{ij}} \tag{3}$$

The E_{loc} quantifies the efficiency of information exchange within the neighborhood of a given cortical region, reflecting local integration capacity. During fine motor task execution, higher E_{loc} indicates that adjacent regions can effectively compensate and maintain stable information processing.

$$E_{loc}(v) = \frac{1}{N}\sum_{v}\frac{1}{k_v(k_v-1)}\sum_{i,j\in N(v)}\frac{1}{d_{ij}} \tag{4}$$

where $N(v)$ denotes the set of nodes functionally connected to node v. C_d indicates the degree of connectivity between brain regions during task performance. Higher complexity actions typically require greater multisite sensorimotor coordination and thus exhibit increased network integration and stronger functional connectivity patterns.

$$C_d = \sum_{i,j\in N(v)} W(i,j) \tag{5}$$

where $W(i,j)$ denotes the weight of the edge between nodes i and j.

It is important to note that graph-theoretical network metrics were used solely for neuroscientific interpretation and were not incorporated as features in the classification model.

2.6. Complexity recognition for fine-motor tasks

2.6.1. Label Assignment

Each of the five fine-motor tasks was assigned a label from 1 to 5, as illustrated in Fig. 4. The fNIRS time-series data were segmented into fixed-length intervals using a sliding window approach. A sliding window of 200 time points (4 s at a 50 Hz sampling rate) with a step size of 25 time points (0.5 s) was applied to segment the fNIRS time-series data. This configuration balances temporal completeness and information density, enabling effective capture of task-related hemodynamic responses while limiting redundancy between adjacent segments.

2.6.2. Construction of the attention-based Bi-LSTM model

An attention-based Bi-LSTM model was developed to classify the five fine-motor tasks [36,37]. The network consists of three parts, namely, a Bi-LSTM layer, an Attention mechanism, and a fully connected output layer. The Bi-LSTM layer captures temporal dependencies in the fNIRS time-series data by integrating contextual information from both forward and backward directions. The attention mechanism computes dynamic weights over the temporal dimension, enabling the model to focus selectively on segments that carry greater task-relevant information. Finally, the aggregated features are passed through a fully connected layer to generate class probabilities, enabling accurate task classification across varying complexity levels. The framework of the proposed detection method is shown in Fig. 5.

The following provides a detailed description of the three core components. First, LSTM is a type of recurrent neural network extensively applied in medical signal processing tasks, including disease diagnosis and physiological time-series analysis [38–40]. Its architecture is designed to capture long-range dependencies by maintaining two internal states, i.e., the cell state (c^t) and the hidden state (h^t), where t denotes the time step. Information flow within the LSTM is modulated through three gating mechanisms: the forgetting gate, the input gate, and the output gate. The forgetting gate determines the information (Z^f) that should be retained from the hidden state (h^{t-1}) of the moment previous to the current moment.

$$Z^f = \delta(W_f[h^{t-1}, x_t] + b_f) \tag{6}$$

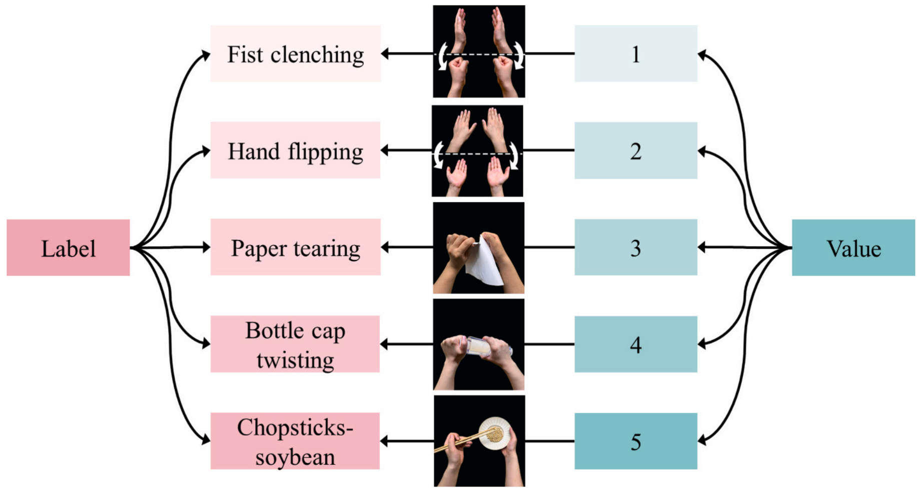

Fig. 4. Label mapping diagram for the five fine-motor tasks.

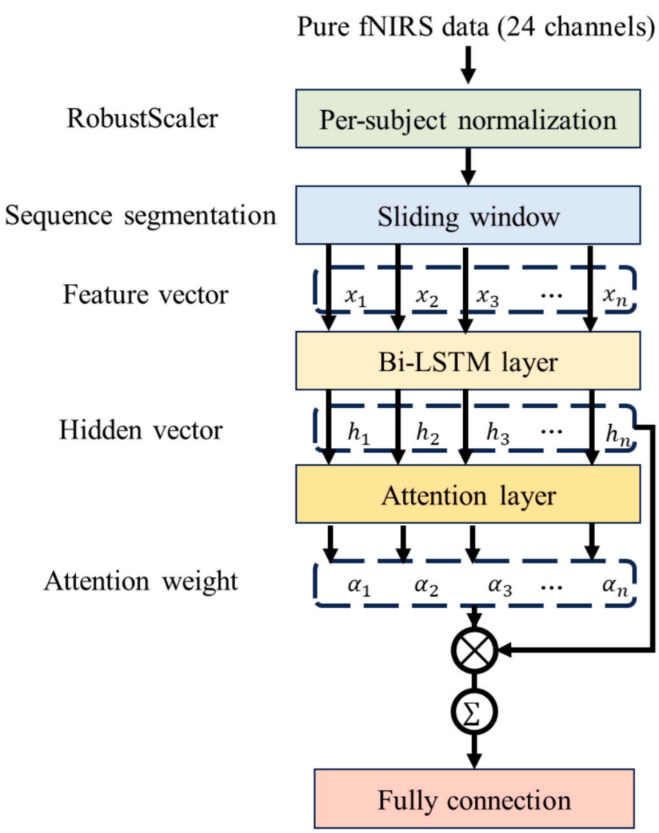

Fig. 5. Architecture of the attention-based Bi-LSTM model.

where x_t is the input at the current step, δ is the activation function, W_f and b_f are the weight and bias of the forgetting gate, respectively. The input gate will be responsible for determining the amount of input information (Z^i) retained in the cell structure.

$$Z^i = \delta(W_i[h^{t-1}, x_t] + b_i) \quad (7)$$

$$\widetilde{c}^t = \tanh(W_c[h^{t-1}, x_t] + b_c) \quad (8)$$

$$c^t = Z^i \times \widetilde{c}^t + Z^f \times c^{t-1} \quad (9)$$

where $\widetilde{c}^t$ is the generated new cell state, c^t is the final retained cell state, W_i and W_c are the weights of the input gate and the new cell state, respectively, and b_i and b_c are the biases. The output gate determines the amount of information Z^o output from c^t.

$$Z^o = \delta(W_o[h^{t-1}, x_t] + b_0) \quad (10)$$

$$h_t = Z^o \times \tanh(c_t) \quad (11)$$

where W_o and b_o are the weights and biases of the output gates. Each gate accomplishes the task of forgetting and remembering by regulating the relevant unit information.

Nonetheless, when dealing with long-time sequences, LSTM may face the problem of vanishing or exploding gradients, leading to training difficulties and performance degradation. To address this limitation, we adopt a Bi-LSTM architecture that integrates forward and backward LSTM units, enabling the model to capture both past and future dependencies. In turn, this alleviates the problem of vanishing and exploding gradients.

$$y_t = \delta(W_0[\overrightarrow{h_t}, \overleftarrow{h_t}] + b_h) \quad (12)$$

where $\overrightarrow{h_t}$ is the output of the forward LSTM cell, $\overleftarrow{h_t}$ is the output of the backward LSTM cell, b_h is the bias of the final merged hidden state, and y_t is the output of the Bi-LSTM cell.

To highlight the important information implicit in the smart contract opcode, the attention layer first calculates the feature matrix X_c output from the Bi-LSTM layer to obtain its implicit representation H, and then calculates the similarity between H and the random initialization parameter matrix W^T, normalizes the weight coefficients assigned to the input feature vector in the overall semantic scenario using the softmax function to obtain the weights α, and finally multiplies the weight matrix α with the feature matrix X_c to obtain the implicit features Y of the opcode, as shown in the following equation.

$$H = \tanh(W_a \times X_c + b_a) \quad (13)$$

$$\alpha = softmax(W^T \cdot H) \quad (14)$$

$$C = X_c \times \alpha^T \quad (15)$$

where W_a denotes the attention layer weight matrix and b_a denotes the bias vector of the attention layer.

Finally, after the input data is processed by the Bi-LSTM layer and the attention layer, the fully connected layer is introduced. This serves to map the fused features to the task label space, realizing the classification of five types of fine-motor tasks with different degrees of complexity.

The cross-entropy loss function is subsequently employed to calculate the loss by comparing the predicted logits with the ground-truth labels. This function implicitly integrates a softmax operation, transforming the outputs of the fully connected layer into a normalized probability distribution over the task categories. Additionally, the Adam optimizer was used to update model parameters, combining the benefits of adaptive learning rates and momentum to accelerate convergence and stabilize training. The learning rate was set to 0.001, and L2 regularization (weight_decay = 5e-3) was applied to mitigate overfitting.

2.6.3. Model evaluation

To avoid subject-specific bias arising from random allocation and to enhance the model's generalization ability to unseen individuals, this study employed a Leave-One-Subject-Out (LOSO) evaluation strategy, which is a subject-based implementation of the broader Leave-One-Out Cross-Validation (LOOCV) framework. Additionally, to further improve model robustness and reduce the impact of randomness during training, this study introduced multi-seed training within the LOSO framework. This involves setting three distinct random seeds for each LOSO iteration, enabling independent model training and evaluation. Within each fold of the LOSO cross-validation, the scaling parameters of the RobustScaler were estimated exclusively from the training subjects to define a unified signal reference framework. This normalization framework was then applied to the continuous fNIRS time-series data of both the training subjects and the held-out test subject, followed by sliding-window segmentation.

Final model performance is reported as the mean ± 95% confidence interval (CI) for multi-dimensional metrics, including Accuracy, Precision, Recall, F1-score, and area under the curve (AUC), yielding more stable and reliable evaluation results. During training, an Early Stopping strategy prevents overfitting, while Dropout regularization suppresses feature redundancy and improves model generalization [41]. In addition, the optimized model was evaluated on the basis of both classification accuracy and receiver operating characteristic (ROC) metrics.

3. Results

3.1. Subjective validation of task complexity

To validate the structural complexity of the five fine-motor tasks, a five-point ranking-scale survey was conducted. Rather than evaluating personal execution difficulty, respondents were instructed to judge the intrinsic structural complexity of each task (e.g., hand-eye coordination, precision demand, force exertion, and muscular engagement). A total of 100 external evaluators aged 18–70 participated, including 15 fNIRS subjects, to ensure a diversified life experience and prevent bias from a single age group. The survey data were used only for task ranking and were not incorporated into the machine learning pipeline in any form. The rating results are summarized in Table 1.

To assess differences in perceived complexity across the five fine-motor tasks, a Friedman test was conducted. Rank-based normalization was applied to mitigate individual rating bias. The analysis revealed significant differences in perceived task complexity (Friedman $\chi^2(4) = 188.4$, $p < 0.001$). Fist clenching and the chopsticks-soybean task were most frequently rated as the least and the most complex. More precisely, 63% (95% CI: 53.2–72.1%) of respondents rated fist clenching as the easiest task, and 73% (95% CI: 63.6–81.2%) rated chopsticks-soybean as the most difficult. Paper tearing and bottle cap twisting were consistently rated as medium-complexity by 58% (95% CI: 47.9–67.7%) and 56% (95% CI: 45.8–65.8%) of participants, respectively, indicating strong agreement in perceived task complexity.

Table 1
Rating scores for the five tasks.

Task	Score (Mean ± SE)	1	2	3	4	5	Number
Fist clenching	4.20 ± 0.12	63	14	9	8	6	100
Hand flipping	3.56 ± 0.11	17	47	16	15	5	100
Paper tearing	3.08 ± 0.08	5	19	58	15	3	100
Bottle cap twisting	2.48 ± 0.10	6	10	19	56	9	100
Chopsticks-soybean task	1.69 ± 0.13	9	5	5	8	73	100

Labels 1–5 represent the five fine-motor tasks ordered by increasing complexity (1 = lowest, 5 = highest). The values denote the number of participants selecting each rating, with 100 total respondents.

To further validate the task-complexity classification in behavioral terms, average execution times were analyzed for each of the five fine-motor tasks. Fig. 6 shows a monotonic increase in mean task completion times: 1.09 s, 1.36 s, 2.33 s, 2.87 s, and 3.45 s, corresponding to rising task complexity. This temporal trend aligns with behavioral ratings, reinforcing the validity of the complexity hierarchy.

3.2. fNIRS data analysis

3.2.1. Characterization of cortical activation under tasks of varying complexity

Baseline correction was first applied to the preprocessed HbO signals, yielding average HbO concentration changes across 24 channels for all the participants, as shown in Fig. 7. A one-sample *t*-test was then performed to determine whether task-related activation significantly deviated from the baseline level. To identify channel-specific differences between task and rest conditions, paired-sample *t*-tests were performed for each channel. Moreover, to control for multiple comparisons, a false discovery rate (FDR) correction was applied.

As shown in Fig. 7, most channels exhibited positive mean HbO values, indicating a general increase during task performance. The standard deviation (0.0896 ± 0.0246) reflected moderate variability and stable signal consistency across trials. All five tasks were associated with significant HbO increases relative to rest (one-sample *t*-tests, $p < 0.05$), with channel-wise paired *t*-tests confirming localized effects after FDR correction ($p < 0.05$). All tasks elicited a characteristic hemodynamic response pattern, characterized by an initial rise in HbO followed by a gradual decline. Tasks involving lower complexity-fist clenching and hand flipping-were associated with relatively modest HbO increases in the PMC and M1 cortices, reflecting a lower level of cortical activation. In contrast, higher-complexity tasks, such as paper tearing, bottle cap twisting, and the chopsticks-soybean task, produced markedly greater HbO elevations, indicative of enhanced neural activation. Further to characterize cortical activation across the five fine-motor tasks, topographical brain maps were generated for each task stage.

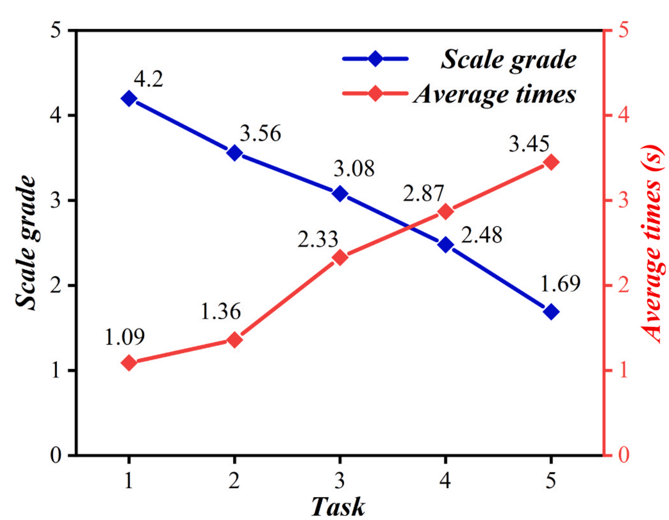

Fig. 6. Complexity ratings and average execution time of different tasks.

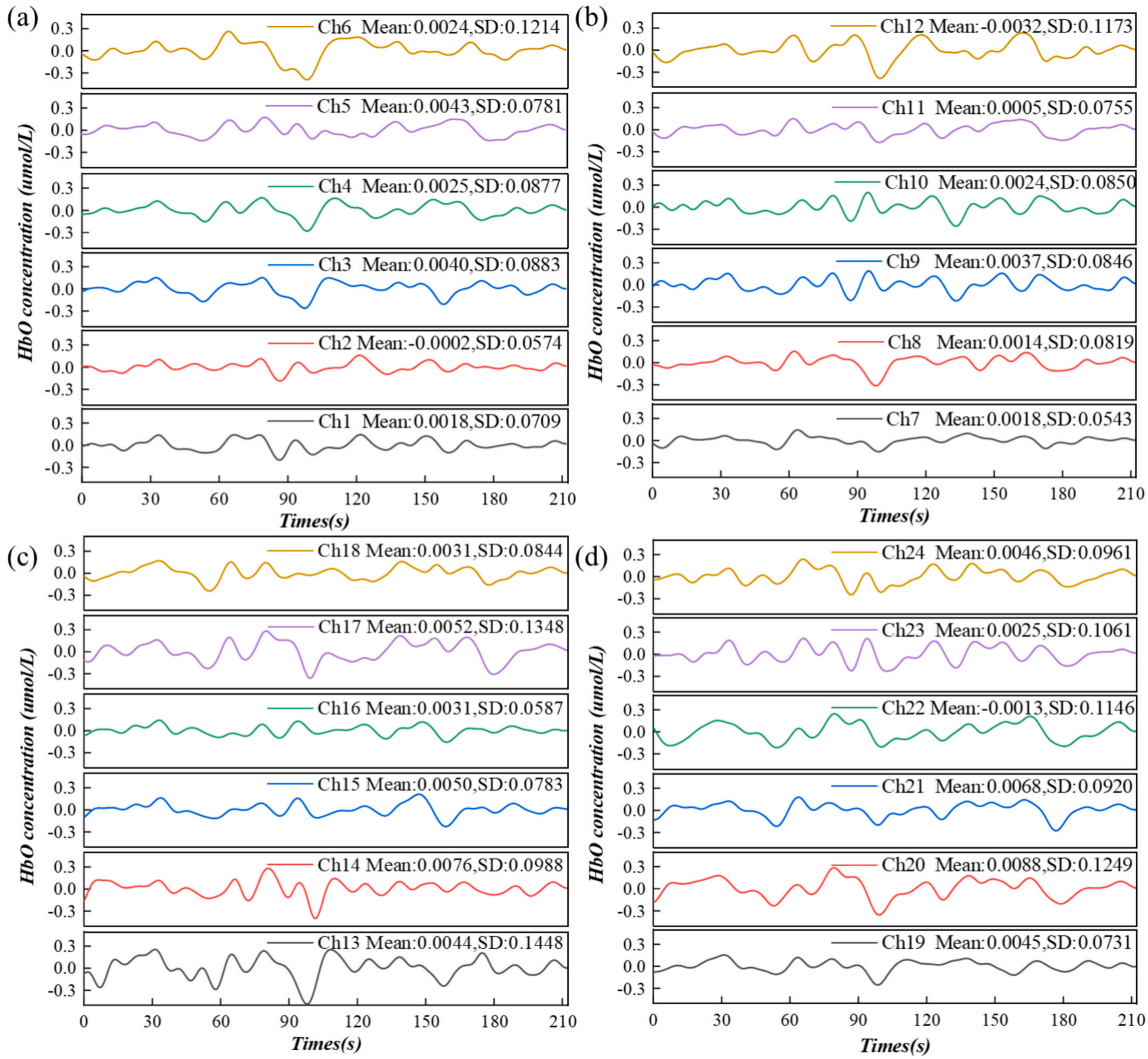

Fig. 7. Mean changes in HbO concentration across all participants for 24 fNIRS channels. (a) Channels 1–6, (b) Channels 7–12, (c) Channels 13–18, (d) Channels 19–24.

As Fig. 8 illustrates, five tasks elicited stronger activation in the right PMC and M1 relative to the left, indicating lateralized engagement of the right hemisphere in bilateral motor control. Moreover, the extent and intensity of cortical activation were positively correlated with task complexity. Lower-complexity tasks such as fist clenching and hand flipping induced only mild, localized activation in the right M1. In contrast, paper tearing and bottle cap twisting evoked significantly stronger activation, particularly in the posterior precentral gyrus and adjacent motor regions, indicating increased M1 involvement in task execution. The chopsticks-soybean task triggered the most extensive activation, with robust engagement of the PMC and M1, consistent with the high precision and coordination demands of this task.

Repeated-measures ANOVA showed a significant main effect of task complexity on HbO amplitude across the five motor tasks ($F = 3.806$, $p = 0.007$, partial $\eta^2 = 0.142$). Subsequent trend analysis demonstrated a significant linear increase in activation strength with rising task complexity ($F = 5.320$, $p = 0.03$, partial $\eta^2 = 0.188$).

To rule out potential cumulative effects from task order, a sequence-effect control analysis was conducted on the activation data. Results indicate that after controlling for the sequence factor, the distinct activation patterns between tasks of varying complexity remained consistent, suggesting that the primary effects were not dominated by task order.

3.2.2. Topological analysis of brain networks under tasks of varying complexity

To investigate how functional brain connectivity varies with task complexity, Pearson correlation coefficients were computed between the HbO time series of all channel pairs. The results were used to construct the average functional connectivity networks for each task condition, with the connectivity patterns shown in Fig. 9.

Both functional connectivity strength and inter-regional

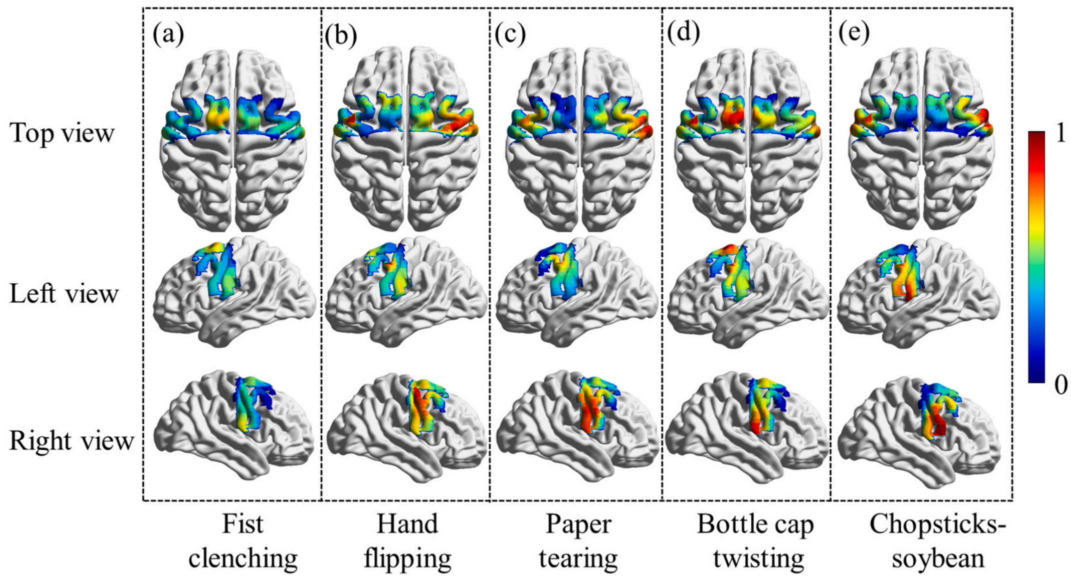

Fig. 8. Top-, left-, and right-view average brain activation maps across all participants for the five fine-motor tasks. (a) Fist clenching; (b) Hand flipping; (c) Paper tearing; (d) Bottle cap twisting; (e) Chopsticks-soybean task.

coordination increased progressively with task complexity, highlighting the adaptive reorganization of the brain under varying motor demands. In low-complexity tasks (fist clenching and hand flipping), PMC-M1 connectivity was weak (mean $\approx$ 0.4). Intra- and inter-hemispheric couplings were relatively limited, as reflected in the dominant blue lines in Fig. 9a, 9b. In contrast, the paper tearing and bottle cap twisting tasks induced broader cortical coordination, with inter-hemispheric connectivity significantly stronger than intra-hemispheric links, as in Fig. 9c, 9d. Among all the tasks, the chopsticks–soybean condition generated the strongest functional connectivity (mean $\approx$ 0.9), as evidenced by the dominant red lines in the matrix (Fig. 9e), reflecting widespread cortical synchrony in motor-related areas.

A repeated-measures ANOVA, adjusted using the Greenhouse-Geisser correction ($\varepsilon = 0.747$), was conducted to assess differences in functional connectivity strength across the five fine-motor tasks. The analysis revealed a significant main effect of task complexity ($F = 17.260$, $p < 0.001$, partial $\eta^2 = 0.429$). Bonferroni-adjusted post hoc analysis identified the greatest difference in connectivity strength as occurring between fist clenching and the chopsticks-soybean task ($p < 0.001$). Connectivity strength did not differ significantly between the paper tearing and bottle cap twisting tasks ($p = 0.09$), indicating similar levels of inter-regional coordination.

Post hoc power analyses were performed based on the main effects of task complexity on HbO activation and functional connectivity. The resulting statistical power ($1 - \beta$) was approximately 0.94 for HbO activation and 0.99 for functional connectivity, indicating that, under the current sample size, the study has adequate sensitivity to detect complexity-related differences in cortical activation and network connectivity.

Further to quantify the topological structure of brain networks, graph-theoretical analysis was applied to the functional connectivity matrices. In light of the small-world properties typically exhibited by cortical networks, connectivity matrices were binarized using an absolute thresholding range of 0.1 to 0.5 (step size = 0.05). Four key topo-

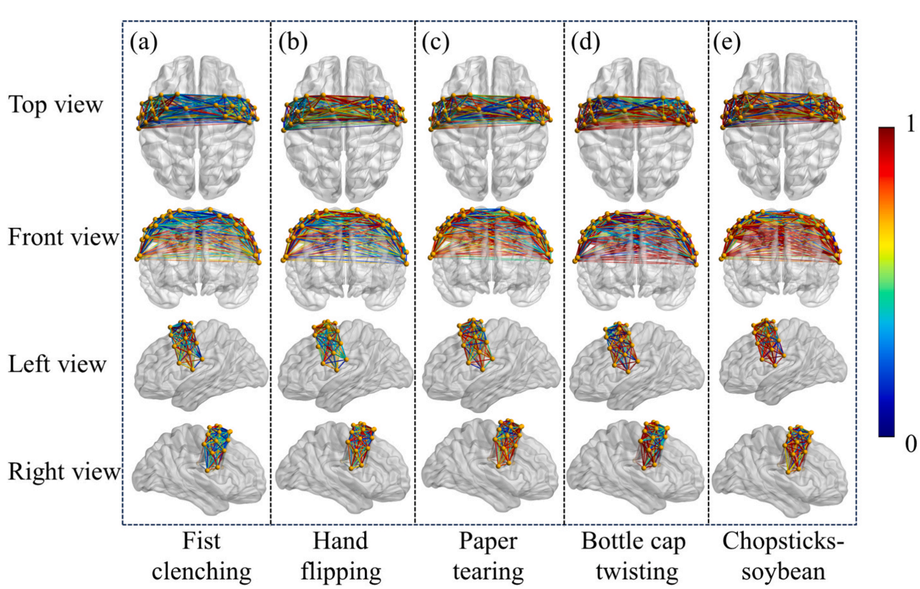

Fig. 9. Mean functional connectivity maps from top, front, left, and right views across all participants for the five fine-motor tasks. (a) Fist clenching; (b) Hand flipping; (c) Paper tearing; (d) Bottle cap twisting; (e) Chopsticks-soybean task.

logical metrics were computed for each task condition, namely, C_p, L_p, E_g and C_d, as shown in Fig. 10.

Among all the tested thresholds, the network topology threshold of 0.25 was the most consistent and recognizable across tasks. This threshold was selected for further analysis due to its optimal stability against complexity-related changes in connectivity. At this optimal threshold, five representative graph-theoretical metrics were computed for each task, C_p, L_p, E_g, E_{loc} and C_d, as illustrated in Fig. 11.

Fig. 11a shows the mean values and variability of graph-theoretical metrics across the five motor tasks. The C_p and E_{loc} exhibited consistently high values with minimal variation across tasks. This indicated, stable local connectivity patterns irrespective of task complexity. In contrast, L_p and E_{loc} exhibited greater variability across tasks, suggesting that the global information transfer capacity of the brain was more sensitive to task complexity. Fig. 11 b-f illustrates the complexity-dependent trends of each metric. C_p,E_g,E_{loc} and C_d, all increased with rising task complexity; this suggested that more demanding tasks elicited stronger local clustering, together with enhanced global integration within the cortical network. Conversely, L_p decreased with task complexity, indicating more efficient global information transmission during higher-demand tasks. Notably, the paper tearing task exhibited elevated E_{loc}, comparable to that of the bottle cap twisting task-both classified as medium-complexity. This subtle difference underscores the similarity in local network properties when processing tasks of comparable complexity.

3.3. Analysis of classification results and model evaluation for the attention-based Bi-LSTM

The proposed attention-based Bi-LSTM model was evaluated against several traditional machine learning algorithms and deep learning architectures, with classification results summarized in Table 2. It is important to note that studies referenced in Table 2 differ from the present work in terms of task complexity, data acquisition settings, and participant characteristics. As these factors were not uniformly controlled, performance metrics such as accuracy are only approximately comparable.

The attention-based bidirectional LSTM model proposed by our research achieved an average classification accuracy of 90.67% ± 7.07% under the leave-one-out cross-validation framework, representing an improvement of approximately 9.37–23.78% over conventional methods. These results demonstrate that the model effectively captures the inherent temporal dynamics of fNIRS time-series data.

To further examine the model's capacity to distinguish fine motor tasks at different levels of complexity, task-wise classification accuracy was evaluated using the LOSO cross-validation scheme to ensure subject-independent generalization. Comparative results with SVM and Bi-LSTM models are presented in Fig. 12.

To systematically evaluate classification performance, this study compared the proposed attention-based Bi-LSTM model with traditional SVM and Bi-LSTM models under the same dataset and LOSO cross-validation framework. The results demonstrated that the attention-based Bi-LSTM achieved the highest average accuracy of 90.67% ±

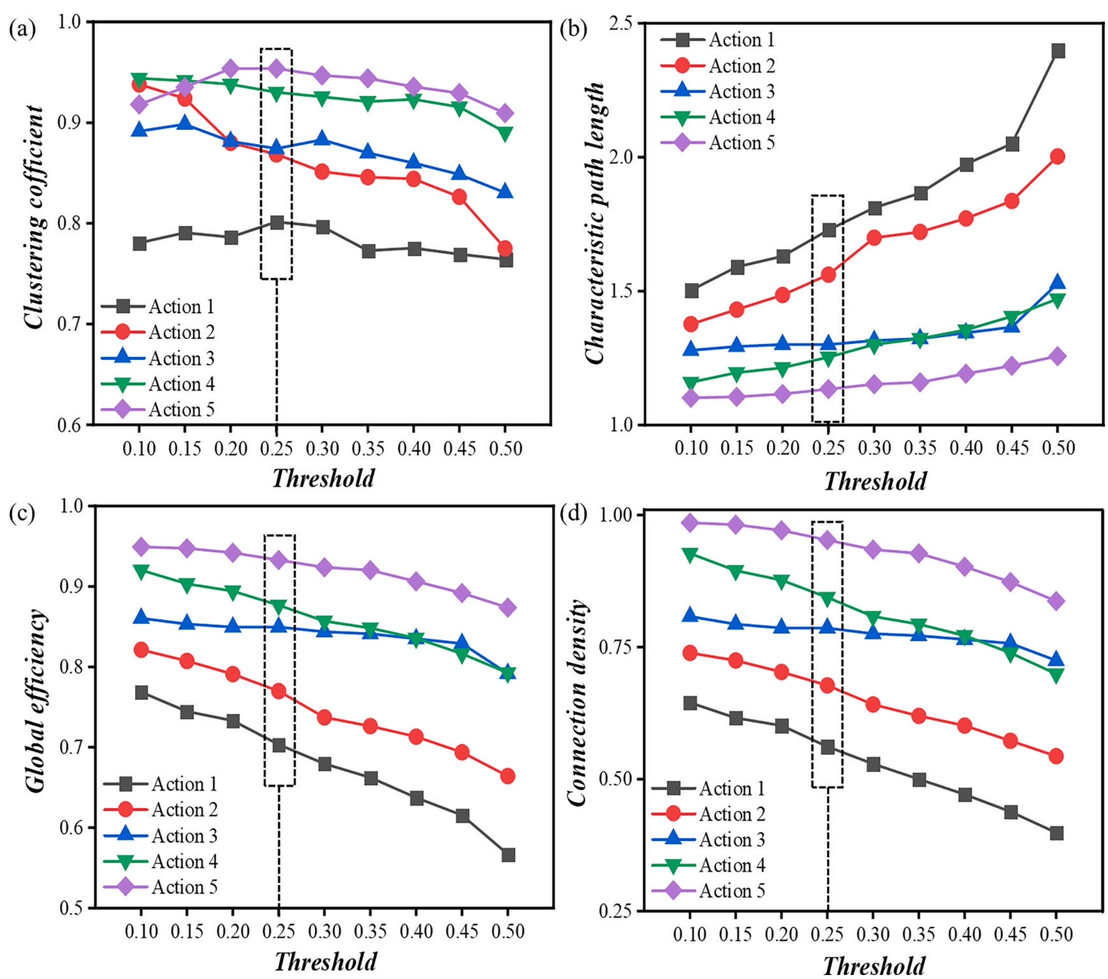

Fig. 10. Comparison of graph-theoretical network metrics across the five fine-motor tasks under absolute thresholds ranging from 0.1 to 0.5. (a) C_p, (b) L_p, (c) E_g, (d)C_d.

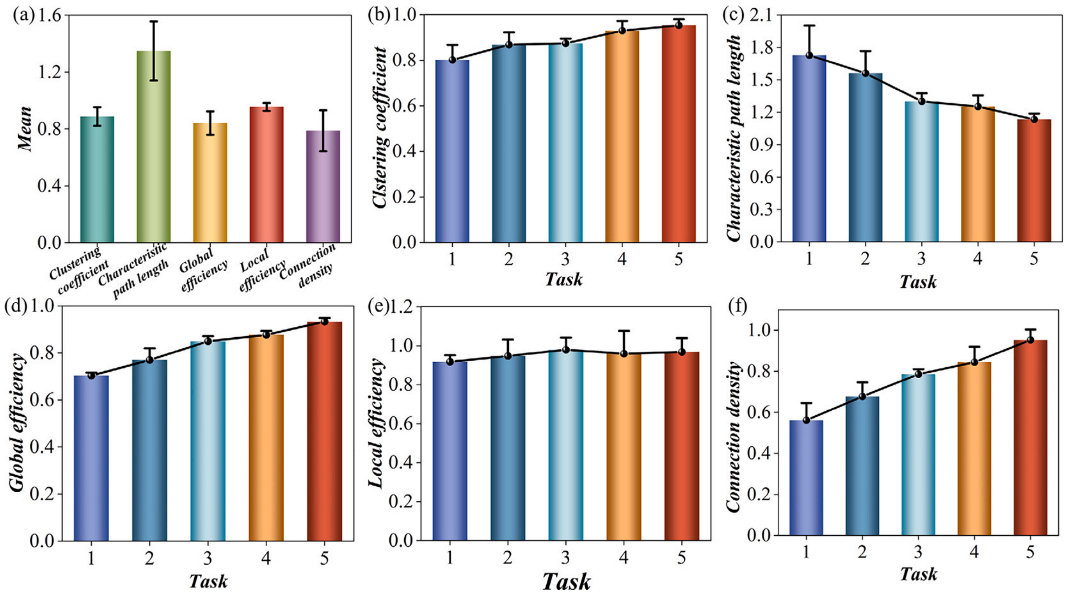

Fig. 11. Graph-theoretical analysis of brain network topology for the five fine-motor tasks at a threshold of 0.25. (a) Mean values of graph-theoretical metrics for the five fine-motor functions, (b) C_p, (c) L_p, (d) E_g, (e) E_{loc}, (f)C_d.

Table 2
Comparison of classification results of different task recognition models.

Researchers	Task	Model	Accuracy ± SE
SuJin et al [47]	Finger&Foot-tapping	SVM + LOOCV	70.4% ± 18.4%
Androu et al [48]	Imagine Playing Tennis	SVM + LOOCV, LDA + LOOCV	76% ± 19.68% 75% ± 20.7%
Lisa et al [49]	Imagine finger-tapping tasks	FLDA + LOOCV	81.3% ± 7.0%
Wang et al [50]	Three types of tapping movements	Transformer + LOSO, CNN + LOSO, LSTM + LOSO	78.22% ± 16.69%, 73.47% ± 17.11%, 66.89% ± 14.29%
This study	Five types of fine hand movements	Attention-based Bi-LSTM + LOSO	90.67% ± 7.07%

7.07% (95% CI: ± 2.68%), clearly outperforming SVM (69.37%) and Bi-LSTM (79.7%). Our model exhibited superior robustness in key evaluation metrics, indicating not only higher overall accuracy but also improved sensitivity and discriminative capability across classes. The evaluation metrics for the other two models are shown in Table 3. Furthermore, the normalized confusion matrix confirms that the proposed model consistently yields higher classification accuracy across all five fine-motor task categories. These results underscore the advantage of integrating attention mechanisms with Bi-LSTM in capturing the temporal dynamics and discriminative patterns of fNIRS-based fine

Table 3
Comparison of classification performance across models under LOSO evaluation.

Model	Accuracy (Mean ± 95% CI)	Precision (Mean ± 95% CI)	Recall (Mean ± 95% CI)	F1-score (Mean ± 95% CI)
SVM	0.6937 ± 0.1742	0.7109 ± 0.1745	0.6937 ± 0.1742	0.6906 ± 0.1743
Bi-LSTM	0.797 ± 0.1761	0.802 ± 0.1984	0.7970 ± 0.1761	0.7968 ± 0.1937
This model	0.9067 ± 0.0268	0.9129 ± 0.0331	0.9067 ± 0.0268	0.8994 ± 0.0316

The reported recall corresponds to the micro-recall across classes.

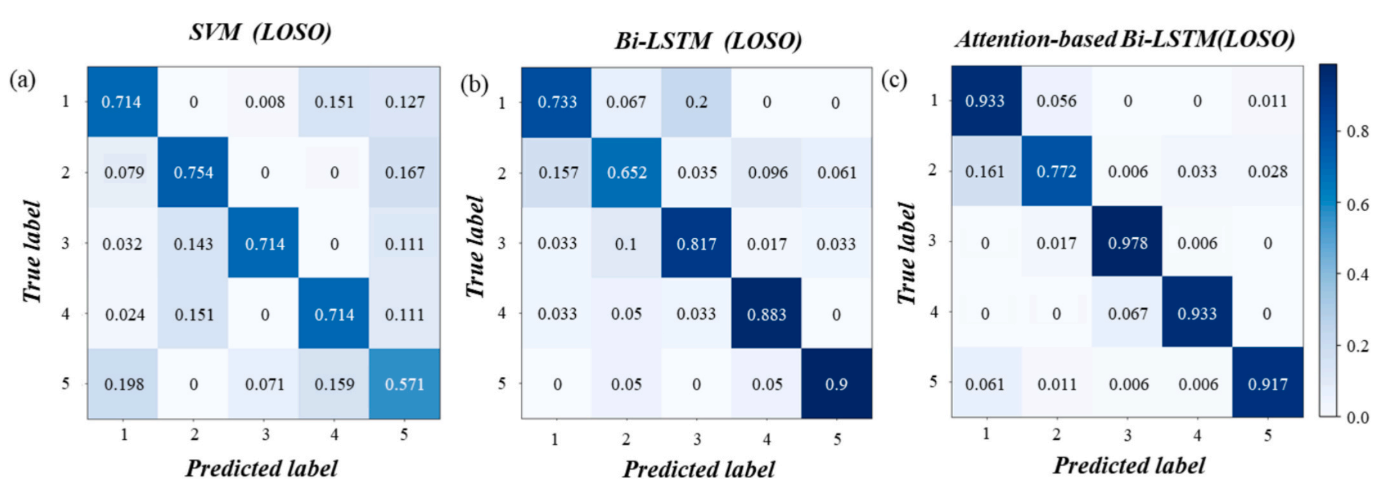

Fig. 12. Confusion matrix showing classification accuracy across the five fine-motor tasks. (a) SVM(LOSO), (b)Bi-LSTM(LOSO), (c) Attention-based Bi-LSTM(LOSO).

motor tasks.

Further to evaluate the classification performance and generalization capability of the proposed model, ROC curves and corresponding area AUC values were computed for each of the five fine-motor tasks. These metrics provide a comprehensive assessment of the sensitivity and specificity of the model across task categories. The results are presented in Fig. 13.

As shown in Fig. 13, the ROC curves for all five tasks were predominantly located in the upper-left quadrant, indicating high true-positive and low false-positive rates. All five tasks achieved AUC values above 0.96, demonstrating strong sensitivity and specificity. Notably, the bottle cap-twisting and chopsticks-soybean tasks achieved AUC values close to 1.0, indicating excellent discriminative performance for higher-complexity tasks.

4. Discussion

4.1. Neurophysiological impact of fine motor task complexity on cortical activation patterns.

This study employed fNIRS combined with graph-theoretical analysis to investigate how fine-motor tasks of varying complexity modulate cortical dynamics within the M1 and PMC. The results indicated that both the spatial extent and the intensity of cortical activation significantly increased with task complexity. This finding aligns with previous fNIRS research demonstrating that complex hand tasks, such as those employed in Fugl-Meyer assessments, can markedly elevate HbO signal amplitude in M1[19]. This reflected greater demands on integrated motor processes such as motor planning, visuospatial coordination, and cognitive control. At the network level, graph-theoretical analysis using a threshold of 0.25 revealed a strong positive correlation between task complexity and global network metrics. Specifically, C_p,E_g,E_{loc} and C_d each increased with task complexity, a pattern consistent with recent hyperscanning research showing that decreased network modularity during successful teacher-child interactions corresponds to increased E_g in prefrontal-temporal-parietal circuits[42], indicating enhanced local segregation and improved global integration to support more demanding motor tasks. Conversely, L_p decreased with increasing task complexity, suggesting more efficient information transfer via shorter communication paths between cortical regions.

Despite increasing task complexity, E_{loc} remained statistically similar between the paper tearing and bottle cap twisting tasks ($\Delta E_{loc} = 0.015$, $p = 0.09$). When viewed in conjunction with execution time and subjective ratings, this finding suggests that both tasks may engage overlapping modules, such as motor sequencing and tool-use coordination. In turn, this demonstrates a possible plateau in the current complexity gradient [43]. Under high-complexity tasks, increased E_g coupled with reduced L_p, may reflect a dual neural strategy, combining rapid information routing with redundant buffering, thereby optimizing behavioral performance. This finding differs from studies of patients with spinal cord injuries [44], indicating that the healthy brain responds to complex challenges through topological optimization rather than simply adding neural resources.

The enhanced cortical activation and more efficient functional network topology of the brain, under high-complexity fine-motor tasks, may serve as potential neural markers for the assessment of individual motor function. In M1 and PMC, the absence of the expected activation or network integration under high-complexity tasks may signal underlying deficits in motor planning or execution. These findings align with previous reports of reduced cortical activation in Parkinson's disease patients, further supporting the potential utility of network-level features for predicting early motor decline [45].

These findings collectively demonstrate that fNIRS-based graph-theoretical analysis serves as a promising non-invasive method for the quantitative assessment of motor function. Importantly, this approach enables not only early screening of neurodegenerative disorders (e.g., detecting aberrant network reorganization in prodromal Parkinson's disease through task complexity-dependent graph-theoretical analysis) but also provides objective quantification of motor function recovery in stroke rehabilitation by monitoring dynamic restoration of functional network topology. Compared with the subjective limitations inherent in the conventional Fugl-Meyer Assessment, this graph-theoretical evaluation strategy provides more sensitive biomarkers for clinical monitoring.

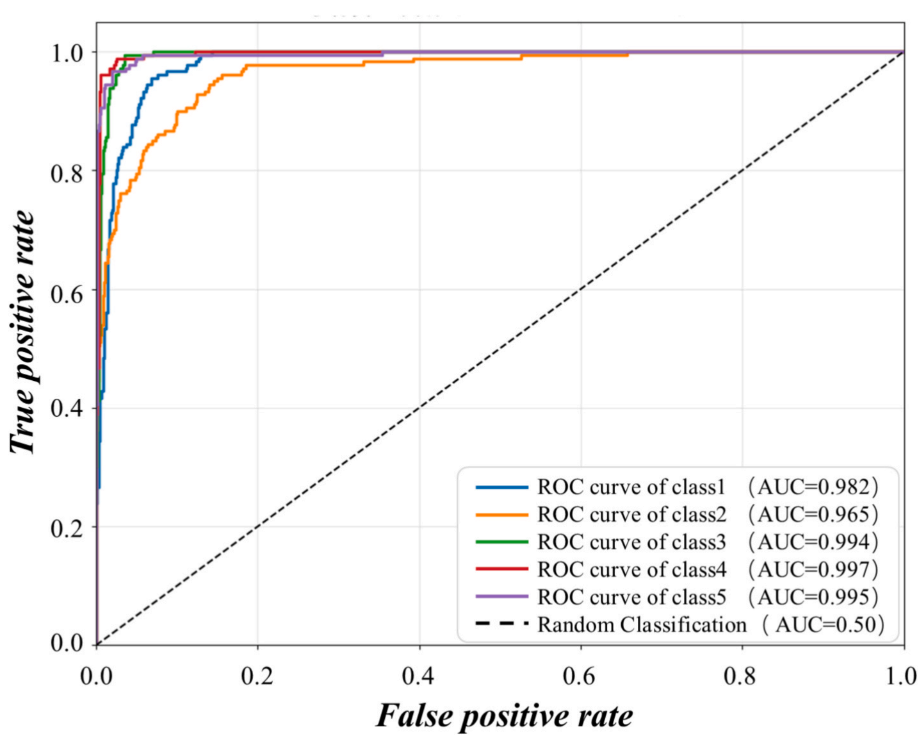

Fig. 13. ROC curves for the attention-based Bi-LSTM model across the five fine-motor tasks.

4.2. Impact of fine motor task complexity classification on BCI strategies

This study presents a sliding-window, attention-based Bi-LSTM model that classifies fine-motor tasks of varying complexity using temporal features extracted from fNIRS signals. The classification results quantitatively revealed the relationship between task complexity and the separability of neural response patterns, providing theoretical foundations for developing adaptive decoding strategies in BCI systems.

The model demonstrates outstanding performance, achieving an accuracy of 90.67% across five categories after LOSO cross-validation. Its average accuracy surpasses that of SVM by 21.3% and that of Bi-LSTM by 10.97%. These results underscore the superior capacity of the model to capture the temporal dynamics of fNIRS signals, particularly in complex motor tasks. This finding suggests that real-time monitoring of neural activity could enable adaptive adjustment of decoder parameters in BCI systems to match varying task demands.

Classification accuracy was, however, comparatively lower for the fist clenching and hand flipping tasks, which exhibited overlapping features, both behavioral and neural characteristics. This convergence stems from their similar motor planning and manipulation requirements, underscoring the potential advantages of incorporating task-context priors to improve task separability within BCI frameworks.

The high accuracy of the model in distinguishing motor tasks across a high-complexity gradient provides a foundation for adaptive BCI design. It enables dynamic adjustment of feature extraction and decision-making processes based on the user's inferred motor intent, thereby enhancing control precision and reducing task misclassification. Such capabilities are particularly valuable in contexts such as neuro-rehabilitation or prosthetic control, where recognizing whether a user is executing a low- or high-complexity task can guide decisions on feedback intensity, assistance levels, or safety constraints.

4.3. Limitations and future work

This study established a systematic framework linking the complexity of fine motor tasks to cortical network dynamics, and validated their separability through decoding models. However, certain limitations and areas for future work require further clarification.

The sample size of this study is relatively small, and all participants were healthy young right-handed adults, which limits generalizability and may introduce selection bias, such as no direct assessment of handedness-related lateralization or age-related effects. This homogeneous design was adopted to establish a clean "healthy neural baseline" and to isolate the intrinsic relationship between task complexity and cortical network dynamics. Post hoc power analysis indicated high statistical power for the primary activation and connectivity effects, supporting the robustness of the core findings. Future work will expand sample diversity in both demographic and clinical dimensions, including different ages, handedness types, and patients with Parkinson's disease, to further examine disease sensitivity and stage-specific network phenotypes associated with task complexity.

Second, this study is constrained by the inherent limitations of fNIRS. Because near-infrared light penetrates only superficial cortical layers, the technique cannot assess subcortical structures such as the basal ganglia, thalamus, or cerebellum—regions highly relevant to the neural mechanisms of motor control and Parkinson's disease. Future work will incorporate multimodal approaches, including short-separation regression, physiological parameter monitoring, fNIRS-EEG integration, and registration with fMRI, to enhance neural specificity and spatial coverage. Additionally, although HbO was used as the primary signal due to its higher sensitivity and signal-to-noise ratio, we acknowledge that HbR provides complementary information about neurovascular coupling. Subsequent studies will systematically include both HbO and HbR to obtain a more comprehensive characterization of hemodynamic and metabolic responses.

At the computational level, there remains room for methodological enhancement. While Pearson correlation effectively captures task-related co-fluctuations in cortical activity, it is vulnerable to noise and shared confounding drivers. Future work will consider more robust connectivity estimators, such as partial correlation or Granger causality, to isolate direct functional interactions better. In parallel, although the Bi-LSTM model offers a favorable balance between accuracy and computational efficiency for offline decoding, exploring emerging architectures, including Transformer-based models, may further improve generalization and enable real-time, adaptive brain-computer interface applications.

The primary objective of this study is to develop a wearable BCI prototype with dynamic complexity perception capability. The first step is to identify cortical activation and functional-connectivity biomarkers that show stable associations with fine-motor task complexity, establishing a neural basis for decoding. The second objective is to early detect motor dysfunction in neurodegenerative disorders by assessing execution efficiency during standardized fine-motor tasks. The system will also serve as an adaptive rehabilitation platform, enabling real-time decoding of brain states to dynamically adjust feedback intensity or virtual environment complexity. Prospective pilot studies in healthy adults, clinical rehabilitation settings, and patients with Parkinson's disease or stroke will validate the clinical feasibility and rehabilitative benefits of a "perception–decoding–intervention" closed-loop framework.

5. Conclusion

This study provides a comprehensive framework for understanding and decoding the neural underpinnings of fine motor complexity. By integrating fNIRS-based graph-theoretical analysis with an attention-based Bi-LSTM model, this study has demonstrated that cortical network dynamics are systematically modulated by task difficulty and that these modulations can be accurately classified [46].

Our neurophysiological analysis revealed that increasing fine motor complexity elicits not only enhanced activation in the M1 and PMC but also a significant reorganization of functional network topology. Specifically, this study observed a shift toward a more integrated and efficient network configuration, marked by increased, C_p, E_g, E_{loc} and C_d, alongside a reduced L_p. Critically, it demonstrates that these distinct, complexity-dependent neural activities can be effectively decoded. The proposed attention-based Bi-LSTM model capitalized on these temporal dynamics to achieve a classification accuracy of 90.67% through LOSO cross-validation, confirming that fNIRS data contains robust, classifiable information about fine motor intent.

These findings carry significant implications for both clinical neuroscience and neuroengineering. The identified network metrics serve as potential quantitative biomarkers for assessing motor function, offering a more objective alternative to traditional clinical scales for early diagnosis and rehabilitation monitoring. Furthermore, the high decoding accuracy of our model lays a robust foundation for developing a next-generation, adaptive BCI that can dynamically adjust to a user's motor intent and task difficulty. While this study provides a strong proof of concept, future work should validate these findings in larger, more diverse cohorts, including clinical populations such as individuals with Parkinson's disease. Such research will be crucial for translating this framework from the laboratory to real-world assistive and rehabilitative technologies.

CRediT authorship contribution statement

Xing Ji: Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization. **Zhong Yin:** Writing – review & editing, Supervision, Methodology. **Yifei Bi:** Writing – review & editing, Supervision. **Kaiwei Yu:** Validation, Methodology, Investigation, Data curation. **Yize Li:** Writing – review & editing, Supervision, Methodology. **Jiafa Chen:** Writing – review & editing, Supervision, Methodology. **Dawei Zhang:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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